Muhammad Nabeed Haider

Computer Science Undergraduate

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PROFESSIONAL SUMMARY

Computer Science undergraduate with experience building AI applications, machine learning projects, and deep learning systems. Worked on audio deepfake prevention, LLM-based applications, and production-oriented AI workflows using PyTorch, LangChain, LangGraph, and Python. Seeking an opportunity to contribute as an AI Engineer while expanding practical machine learning expertise.

EDUCATION

Bachelor of Science in Computer Science — 2026
FAST National University of Computer and Emerging Sciences, Islamabad

- CGPA: 3.56 / 4.00
- Relevant Coursework: Generative AI, Machine Learning, MLOps, Natural Language Processing, Statistical Modelling, Cloud Computing, Information Security

EXPERIENCE

AI Lab Demonstrator — Spring 2026
FAST National University of Computer and Emerging Sciences

- Conducted weekly Artificial Intelligence laboratory sessions for approximately 50 undergraduate students.
- Provided guidance on Python programming, search algorithms, machine learning fundamentals, and debugging techniques.
- Assisted with grading programming assignments, evaluating lab performance, and supporting students during practical sessions.
- Collaborated with the course instructor to maintain a consistent and engaging laboratory environment.

PROJECTS

FakeLess — Audio Deepfake Prevention — 2026
Python • PyTorch • SpeechBrain • HiFi-GAN

- Designed and developed an audio protection framework that reduces the effectiveness of neural voice cloning while preserving perceptual speech quality.
- Implemented a deep learning training pipeline using PyTorch with SpeechBrain speaker embeddings and HiFi-GAN-based audio reconstruction.
- Processed large-scale LibriSpeech datasets through automated preprocessing and feature extraction pipelines for model training.
- Evaluated system performance using objective speech-quality and speaker-similarity metrics including PESQ, STOI, and embedding similarity.
- Developed an end-to-end training pipeline for universal adversarial perturbation learning using LibriSpeech and speaker embedding optimization.

SCAR — Multimodal AI Assistant — 2026
Python • FastAPI • Ollama • n8n • RAG • LoRA

- Developed a Jarvis-inspired multimodal AI assistant capable of conversational interaction, image generation, retrieval-augmented reasoning, and local system automation.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using n8n to retrieve and incorporate knowledge from Google Drive documents into AI responses.
- Fine-tuned Stable Diffusion 1.5 using LoRA to support customized image generation alongside local LLM-powered conversational capabilities.
- Integrated external services including weather, sports, and news APIs to provide real-time information within a unified assistant interface.
- Designed a modular architecture enabling future expansion through additional tools, APIs, and multimodal capabilities.

GymBuddy 2026

Python • FastAPI • Ollama • RAG

- Developed backend REST APIs for an AI-powered fitness assistant using FastAPI.
- Integrated locally hosted Ollama language models to provide context-aware fitness recommendations.
- Implemented a Retrieval-Augmented Generation (RAG) workflow to improve response relevance using external knowledge.
- Collaborated within a team to design, test, and demonstrate backend functionality through a lightweight web interface.

Production MLOps Pipeline 2026

Python • FastAPI • Docker • Prometheus • Grafana

- Developed a production-oriented machine learning inference service using FastAPI and Docker.
- Integrated Prometheus metrics and Grafana dashboards to monitor application health and model-serving performance.
- Implemented CI workflows using GitHub Actions to automate testing and deployment tasks.
- Applied modular software architecture and containerization practices to improve deployment reproducibility and maintainability.

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning: PyTorch, NumPy, Torchaudio

Generative AI: Ollama, LangChain, LangGraph, RAG, Stable Diffusion 1.5, LoRA

Audio AI: SpeechBrain, HiFi-GAN, Speaker Embeddings

Data Processing: LibriSpeech, Feature Extraction, Model Evaluation

Tools: Git, Docker, Linux

ACHIEVEMENTS

- AI Lab Demonstrator for the Artificial Intelligence course at FAST-NUCES.
- Developed multiple end-to-end AI and machine learning projects spanning Generative AI, MLOps, and backend systems.
- Hands-on experience with production-oriented machine learning workflows, CI/CD, Docker, monitoring, and LLM-based applications.